

Vendor Performance Data Analytics

Exploratory Data Analysis Report

Tools Used: Python, Pandas, MySQL, Power BI

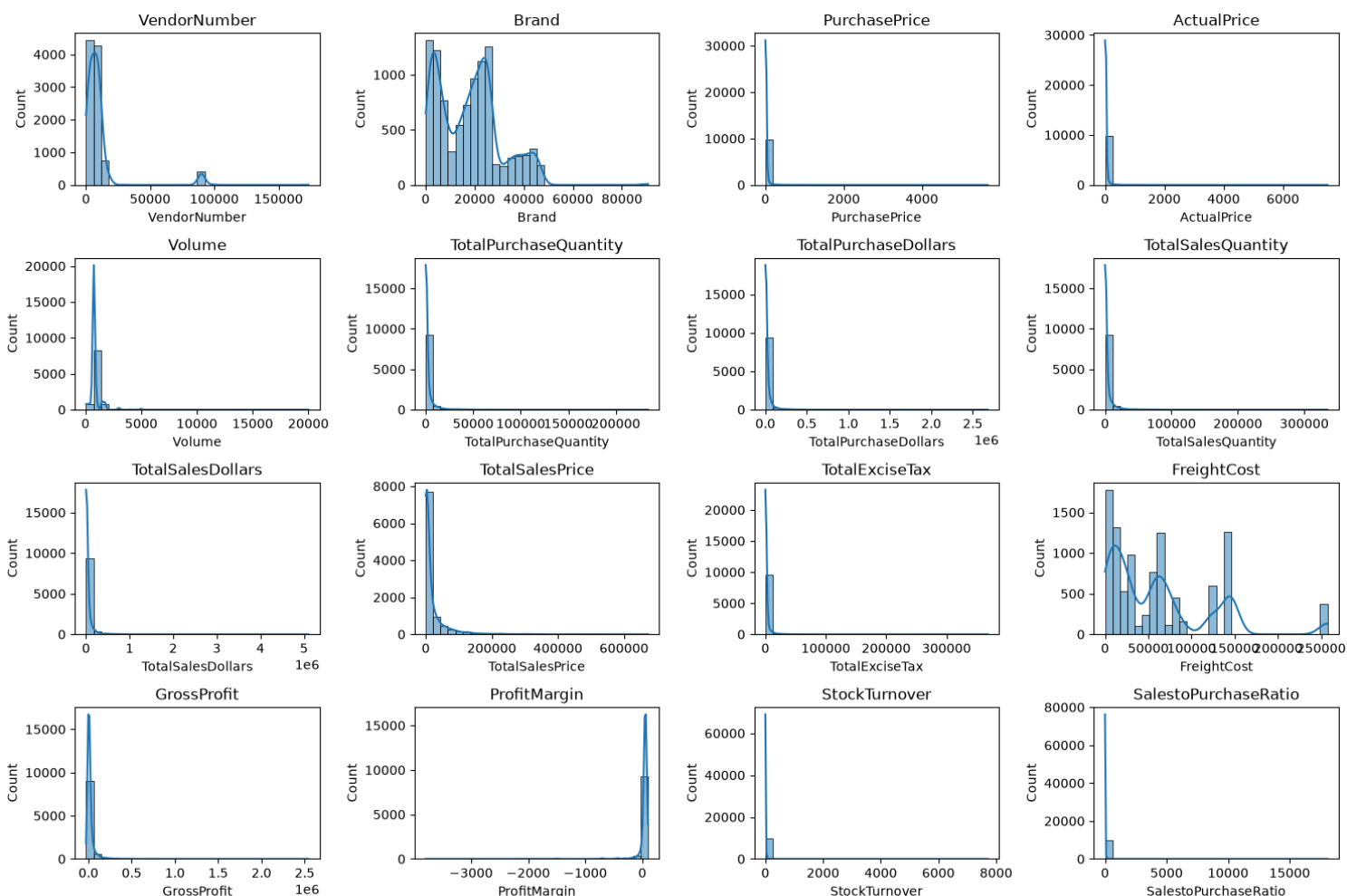
Project Objective:

To analyze vendor sales and purchase performance, identify high-performing and low-performing vendors, evaluate profitability, inventory turnover, and derive actionable business insights.

Exploratory Data Analysis Insights

Summary Statistics

	count	mean	std	min	25%	50%	75%
VendorNumber	9896.0	10411.962308	18204.279316	2.000000	3664.000000	7153.000000	9552.000000
Brand	9896.0	17959.993129	12729.697465	58.000000	5795.750000	18451.000000	25217.000000
PurchasePrice	9896.0	22.879706	106.725152	0.360000	6.770000	10.250000	18.250000
ActualPrice	9896.0	33.605657	145.215352	0.490000	10.490000	15.990000	26.990000
Volume	9896.0	851.238329	652.989043	50.000000	750.000000	750.000000	750.000000
TotalPurchaseQuantity	9896.0	2325.418452	7941.951923	1.000000	26.000000	228.000000	1493.750000
TotalPurchaseDollars	9896.0	21974.575547	87007.479970	0.710000	359.640000	3010.530000	16189.427500
TotalSalesQuantity	9896.0	3305.708064	11348.954981	0.000000	43.000000	336.500000	2192.250000
TotalSalesDollars	9896.0	45399.105508	173859.235113	0.000000	943.610000	6478.955000	32172.462500
TotalSalesPrice	9896.0	20195.089579	46433.182042	0.000000	369.412500	3554.175000	18277.127500
TotalExciseTax	9896.0	1909.719124	11396.436242	0.000000	6.470000	60.655000	484.562500
FreightCost	9896.0	61856.073909	60936.803990	0.270000	14836.570000	55551.820000	79528.990000
GrossProfit	9896.0	23424.529962	89086.281459	-28112.740000	314.985000	3128.125000	16355.575000
ProfitMargin	9896.0	30.953711	118.909187	-3796.044855	34.208509	50.527903	60.590300
StockTurnover	9896.0	4.192158	89.762573	0.000000	1.008497	1.365254	1.652500
SalestoPurchaseRatio	9896.0	6.794281	192.445401	0.000000	1.519953	2.021341	2.537400



Negative & Zero Values:

Gross Profit: Minimum of -52,002.78, indicating potential losses due to high costs or heavy discounts. This could be due to selling products at lower prices than their purchase costs.

Profit Margin: Has a minimum of $-\infty$, which suggests instances where revenue is zero or even lower than the total cost, leading to extreme negative profit margins.

Total Sales Quantity & Sales Dollars: Some products show zero sales, indicating they were purchased but never sold. These may be slow-moving or obsolete stock, leading to inventory inefficiencies.

Outliers Detected by High Standard Deviations:

Purchase & Actual Prices: The maximum values (5,681.81 & 7,499.99) are significantly higher than the mean (24.39 & 35.64), indicating premium product offerings.

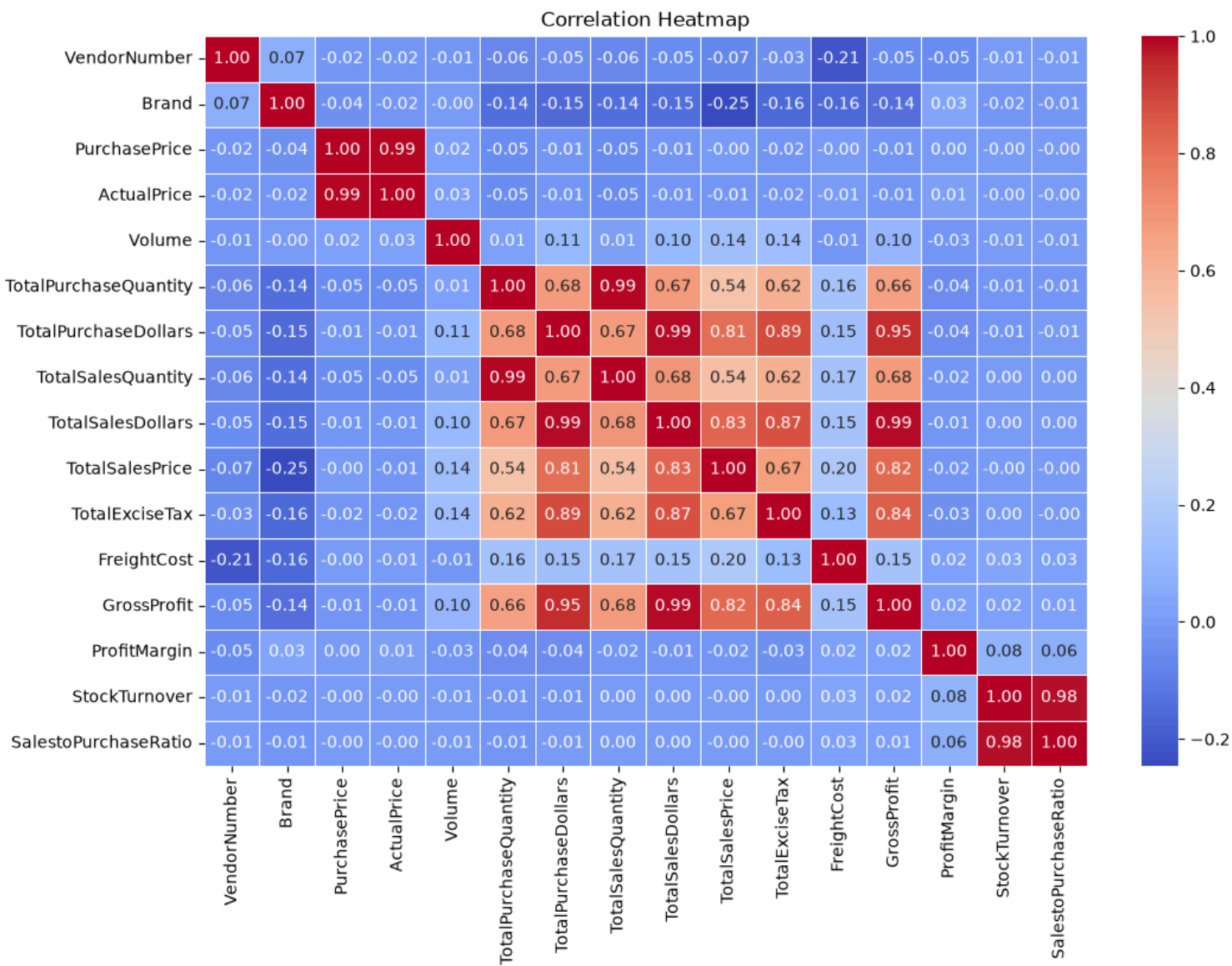
Freight Cost: Extreme variation from 0.09 to 257,032.07 suggests logistics inefficiencies, bulk shipments, or erratic shipping costs across different products.

Stock Turnover: Ranges from 0 to 274.5, suggesting some products sell rapidly while others remain unsold for long periods. A value greater than 1 indicates that sales for a product exceed the purchased quantity due to older stock fulfilling orders.

Data Filtering

To enhance the reliability of the insights, we removed inconsistent data points where:

- **Gross Profit ≤ 0** (to exclude transactions leading to losses).
- **Profit Margin ≤ 0** (to ensure analysis focuses on profitable transactions).
- **Total Sales Quantity = 0** (to eliminate inventory that was never sold).



Purchase Price vs. Total Sales Dollars & Gross Profit: Weak correlation (-0.012 and -0.016), indicating that price variations do not significantly impact sales revenue or profit.

Total Purchase Quantity vs. Total Sales Quantity: Strong correlation (0.999), confirming efficient inventory turnover.

Profit Margin vs. Total Sales Price: Negative correlation (-0.179), suggesting increasing sales prices may lead to reduced margins, possibly due to competitive pricing pressures.

Stock Turnover vs. Gross Profit & Profit Margin: Weak negative correlation (-0.038 & -0.055), indicating that faster stock turnover does not necessarily equate to higher profitability.

Research Questions & Key Findings

1. Brands for Promotional or Pricing Adjustments

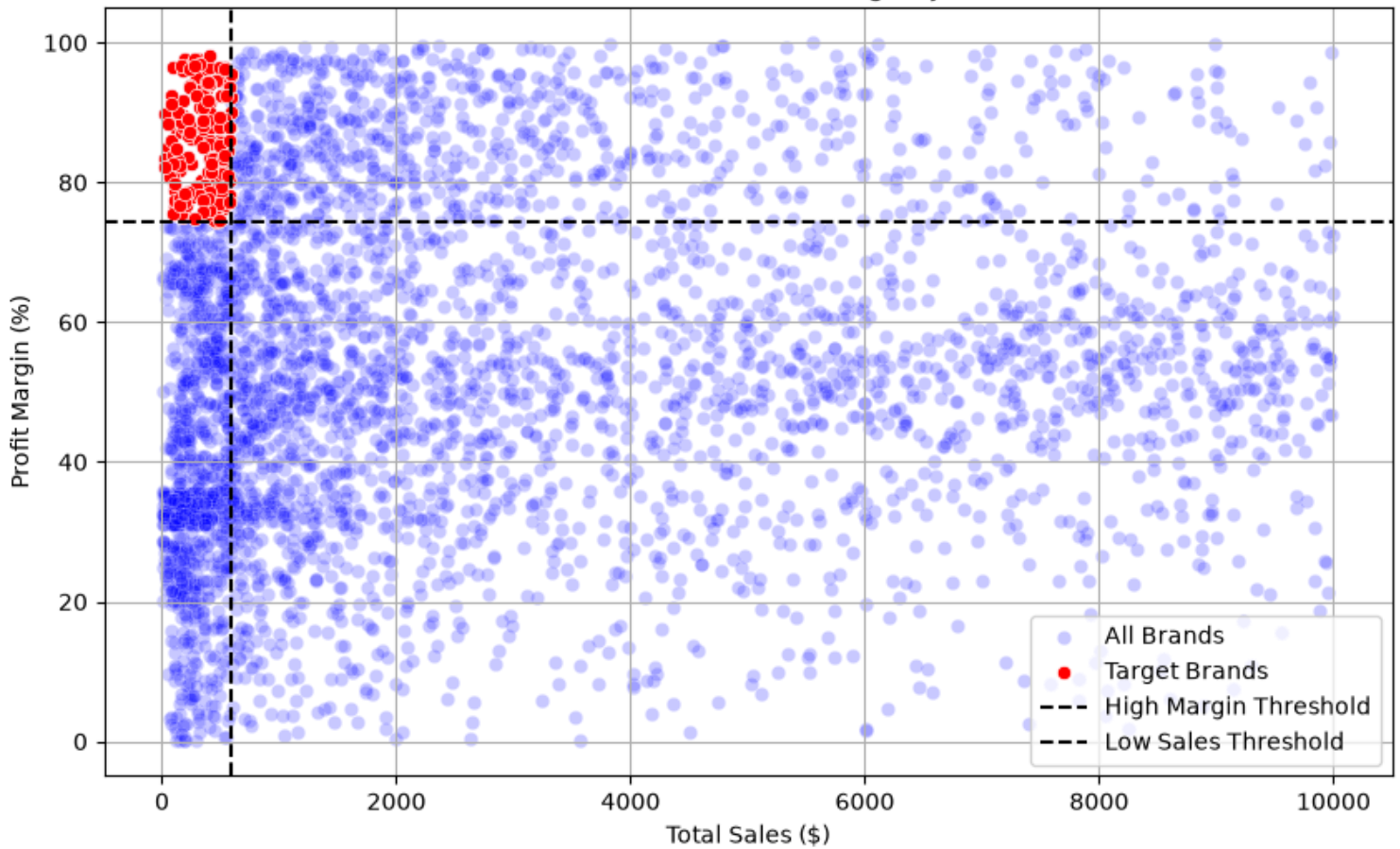
Brands with Low Sales but High Profit Margins:

	Description	TotalSalesDollars	ProfitMargin
2150	Concannon Glen Ellen Wh Zin	15.95	83.448276
2268	Crown Royal Apple	27.86	89.806174
6469	Sauza Sprklg Wild Berry Marg	27.96	82.153076
7461	Tracia Syrah	44.94	88.495772
622	Basilica Amaretto	47.45	85.079031
...	...	...	...
7887	Wyborowa Vodka	579.74	85.935074
1400	Casa Liliana Good Chard	582.47	95.364568
2843	Ermita Sn Felices Rose Rioja	586.63	90.109609
2319	Cupcake Chianti	587.16	92.216772
2228	Courvoisier Rose	591.68	95.436723

171 rows × 3 columns

171 brands exhibit lower sales but higher profit margins, which could benefit from targeted marketing, promotions, or price optimizations to increase volume without compromising profitability.

Brands for Promotional or Pricing Adjustments

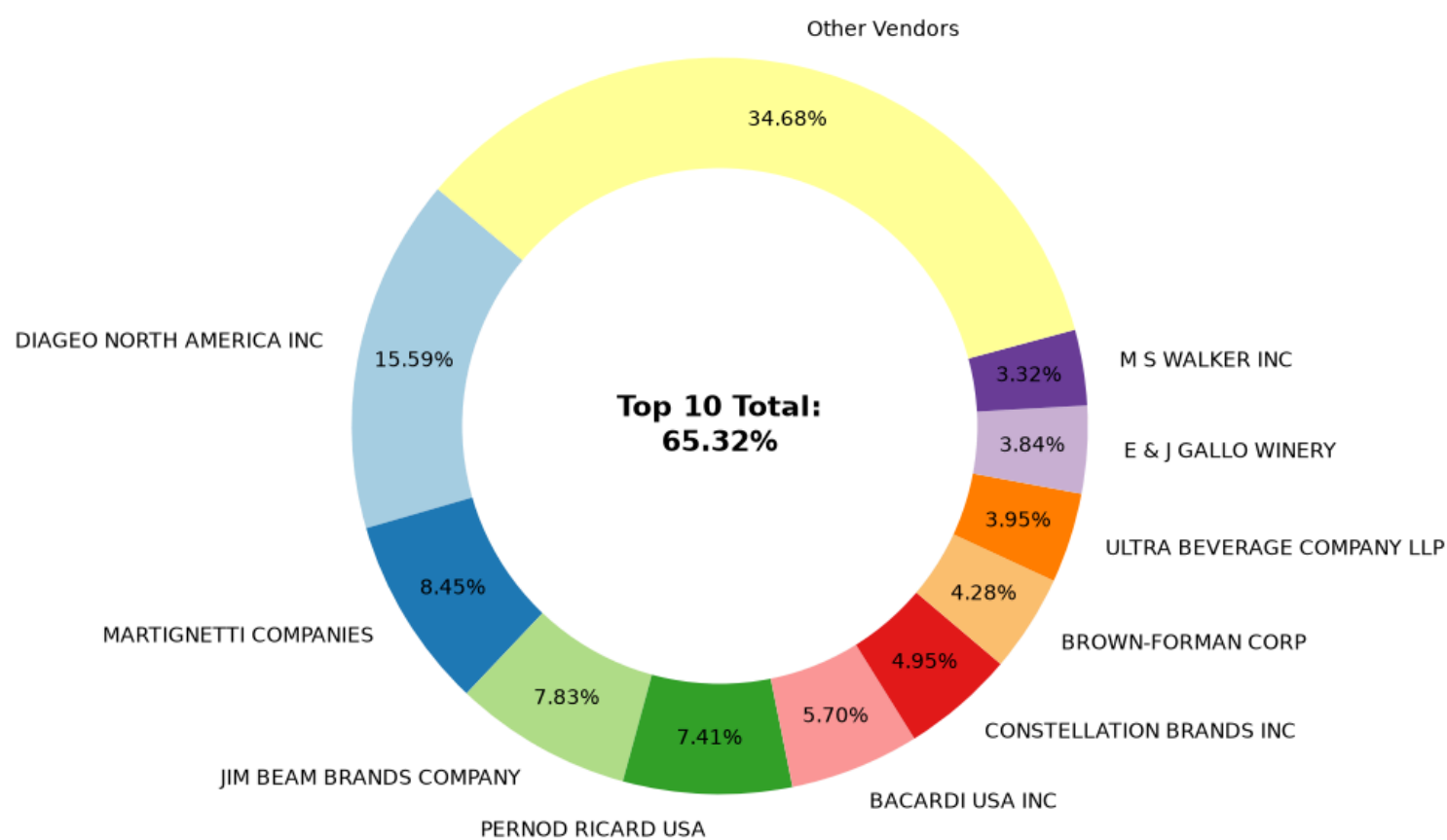


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Top Vendors by Sales & Purchase Contribution

The top 10 vendors contribute **65.32%** of total purchases, while the remaining vendors contribute only **34.68%**. This over-reliance on a few vendors may introduce risks such as supply chain disruptions, indicating a need for diversification.

Top 10 Vendor's Purchase Contribution (%)



Impact of Bulk Purchasing on Cost Savings

Vendors buying in large quantities receive a **70.41% lower unit cost (\$10.75 per unit vs. higher unit costs in smaller orders).**

Bulk pricing strategies encourage larger orders, increasing total sales while maintaining profitability.

UnitPurchasePrice	
OrderSize	
Small	36.337092
Medium	16.147577
Large	10.757976

4. Identifying Vendors with Low Inventory Turnover

Total Unsold Inventory Capital: **\$478.79K**

Slow-moving inventory increases storage costs, reduces cash flow efficiency, and affects overall profitability.

Identifying vendors with low inventory turnover enables better stock management, minimizing financial strain.

	VendorName	UnsoldInventoryValue
34	MARTIGNETTI COMPANIES	68.69K
41	PERFECTA WINES	53.49K
65	ULTRA BEVERAGE COMPANY LLP	49.67K
32	M S WALKER INC	47.61K
28	JIM BEAM BRANDS COMPANY	41.77K
4	BROWN-FORMAN CORP	30.16K
21	FABRIZIA SPIRITS LLC	14.12K
24	FREDERICK WILDMAN & SONS	12.75K
9	CONSTELLATION BRANDS INC	11.83K
18	Dunn Wine Brokers	8.50K

5. Profit Margin Comparison: High vs. Low-Performing Vendors

Top Vendors' Profit Margin (95% CI): (52.01%, 52.95%), Mean: 52.48%

Low Vendors' Profit Margin (95% CI): (48.66%, 50.67%), Mean: 49.66%

Low-performing vendors maintain higher margins but struggle with sales volumes, indicating potential pricing inefficiencies or market reach issues.

Actionable Insights:

- **Top-performing vendors:** Optimize profitability by adjusting pricing, reducing operational costs, or offering bundled promotions.

- **Low-performing vendors:** Improve marketing efforts, optimize pricing strategies, and enhance distribution networks.

Top Vendors 95% CI: (52.01, 52.95), Mean: 52.48

Low Vendors 95% CI: (48.66, 50.67), Mean: 49.66

